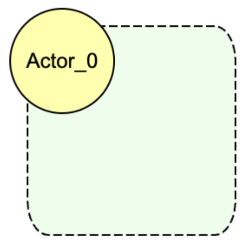
BloomingLeaf User Manual

This document provides a comprehensive guide for BloomingLeaf, containing documentation for all available tools. Users can access assistance with model creation or analysis by clicking on the 'Help' button in the top toolbar, which offers interactive pop-ups featuring instructional videos for each step.

Goal Modeling Refresher:

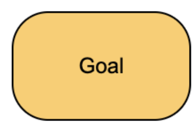
In BloomingLeaf, we use a high-level socio-technical modeling language, meant to answer the questions: who? why? why not? how? how else? and what? It consists of actors, various elements, and links.

Actor: A person, a role, or an organization

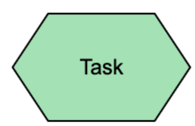


Model Elements

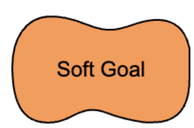
A (hard) **goal:** something clear-cut that an actor wants to achieve.



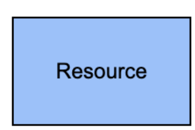
A **task:** an action or step that an actor wants to or must perform.



A **soft goal:** a goal an actor wants to achieve, but of a less clear-cut, fuzzy nature. Usually used to represent qualities or criteria for decisions.



A **resource:** a thing (physical, information, skill) that an actor needs to possess in order to perform a task.

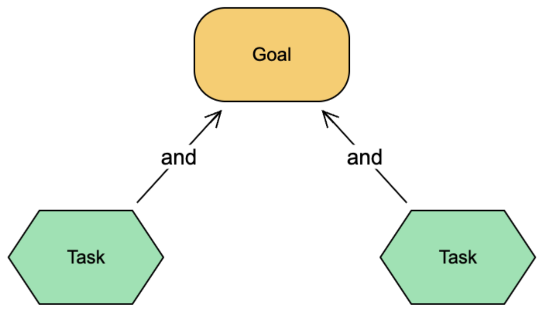


Model Links

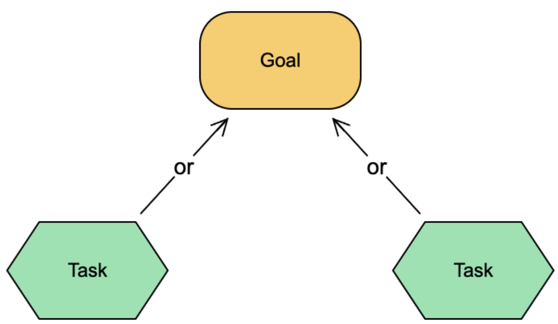
There are two main types of links: decomposition and contributions.

1- Decomposition:

- **And:** to achieve the parent, all the children must be achieved



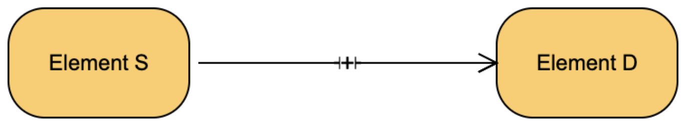
- **Or:** to achieve the parent, one or more alternatives must be achieved



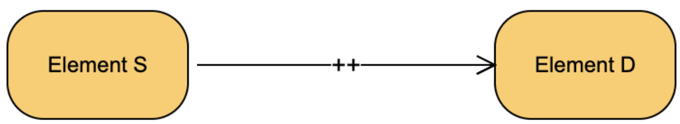
2- Contributions

Contributions can be asymmetric or symmetric:

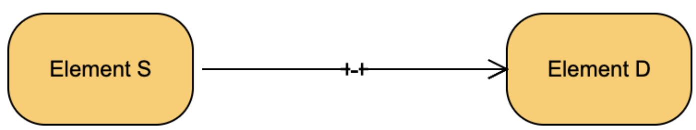
- **Symmetric**
    - +: Evidence for “Element S” propagates some evidence for “Element D” such that 'satisfied' contributes to 'satisfaction' and 'denied' contributes to 'denial.



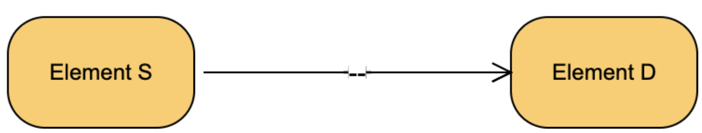
    - ++: Evidence for “Element S” propagates full evidence for “Element D” such that 'satisfied' contributes to 'satisfaction' and 'denied' contributes to 'denial.



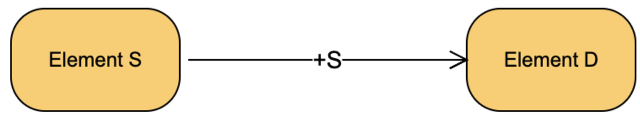
    - -: Evidence for “Element S” propagates some evidence for “Element D” such that 'satisfied' contributes to 'denial' and 'denied' contributes to 'satisfaction’.



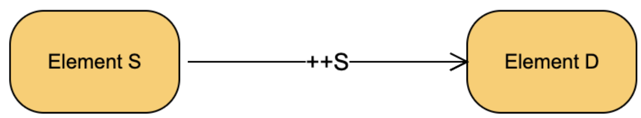
    - --: Evidence for “Element S” propagates full evidence for “Element D” such that 'satisfied' contributes to 'denial' and 'denied' contributes to 'satisfaction’.



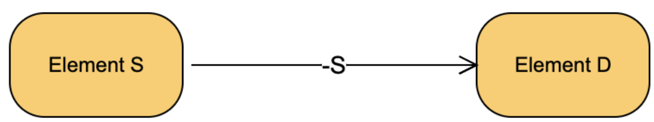
- **Asymmetric:**
    - Satisfied:
        - +S: If “Element S” is satisfied, there is some evidence that “Element D” is satisfied.



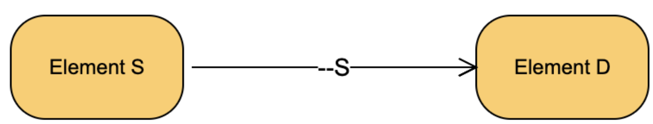
        - ++S: If “Element S” is satisfied, then full evidence exists that “Element D” is satisfied.



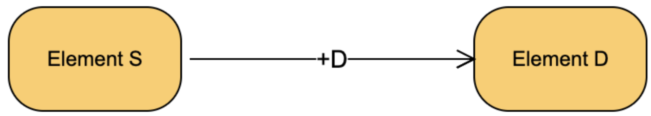
        - -S: If “Element S” is satisfied, there is some evidence that “Element D” is denied.



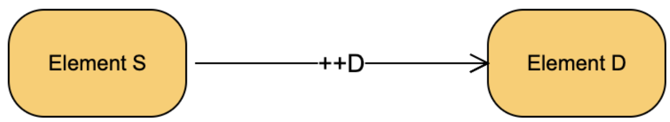
        - --S: If “Element S” is satisfied, then full evidence exists that “Element D” is denied.



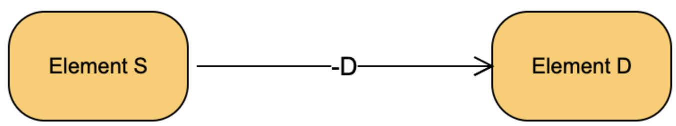
- Denied:
  - +D: If “Element S” is denied, there is some evidence that “Element D” is denied.



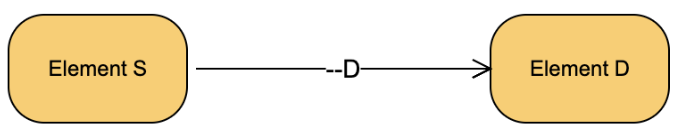
  - ++D: If “Element S” is denied, then full evidence exists that “Element D” is denied.



  - -D: If “Element S” is denied, there is some evidence that “Element D” is satisfied.



  - --D: If “Element S” is denied, then full evidence exists that “Element D” is satisfied.

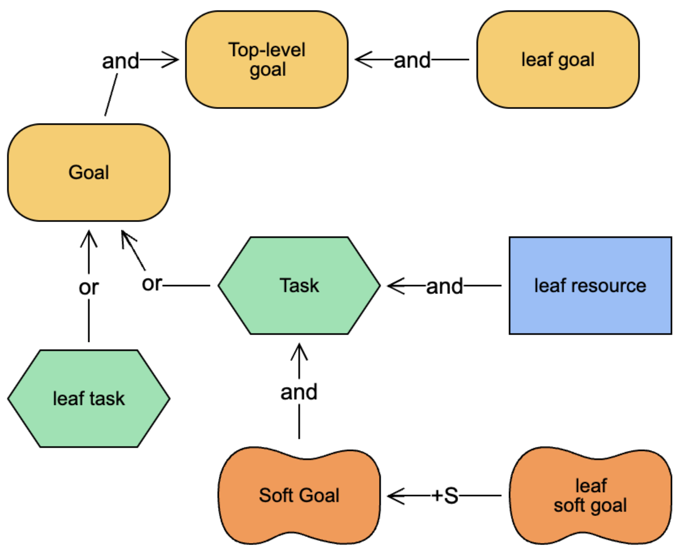


Leaf Nodes and Root (Top-Level) Nodes

Goal models often form tree structures where:

- **Leaf Nodes** are defined as nodes with no incoming decomposition (and/or]) or contribution links (++, -) and NO Outgoing dependency links.
- **Root (Top-Level) Nodes** are defined as nodes with no outgoing decomposition (and/or) or contribution links (++) and NO Incoming dependency links.

Note: A node is a Task, a Goal, A Resource, or a Soft Goal.

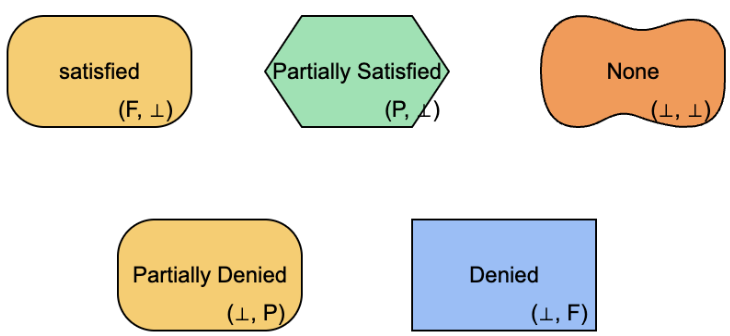


Satisfaction Values

To evaluate the model we use the following satisfaction values

Symbol	Value
(F, ⊥)	Satisfied
(P, ⊥)	Partially Satisfied
(⊥, ⊥)	None
(⊥, P)	Partially Denied
(⊥, F)	Denied
(P, P)	Conflict: Partially Satisfied / Partially Denied
(F, P)	Conflict: Fully Satisfied / Partially Denied
(P, F)	Conflict: Partially Satisfied / Fully Denied
(F, F)	Conflict: Fully Satisfied / Fully Denied

When creating a model, the user can choose to have an Initial Satisfaction Value (Satisfied, Partially Satisfied, None, Partially Denied, or Denied) for each of the different nodes as shown below:

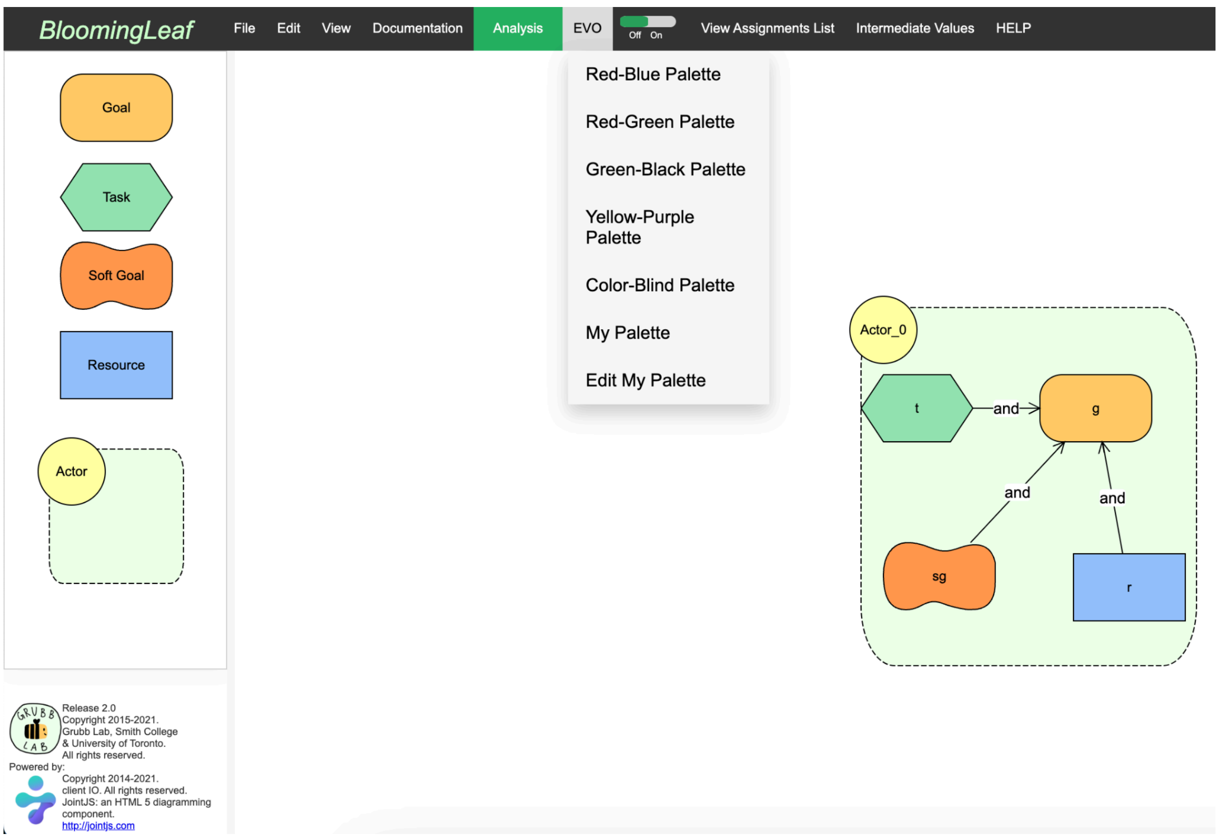


EVO (Evaluation Visualization Overlay)

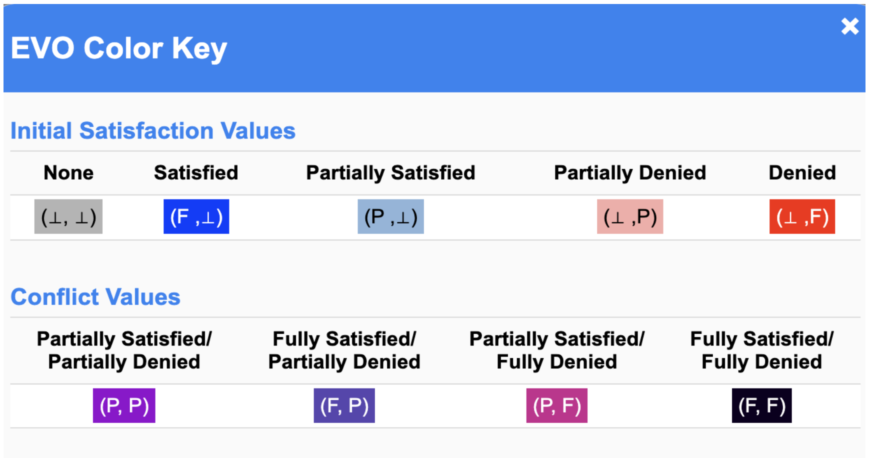
EVO is a feature that facilitates color-coding nodes in the model according to their satisfaction values.

<u>Pre-made Palettes:</u>

The tool provides six different palettes to choose from, along with a color-blind option for individuals with color deficiencies. Simply hovering over the “EVO” button in the top toolbar will reveal all available palette options:

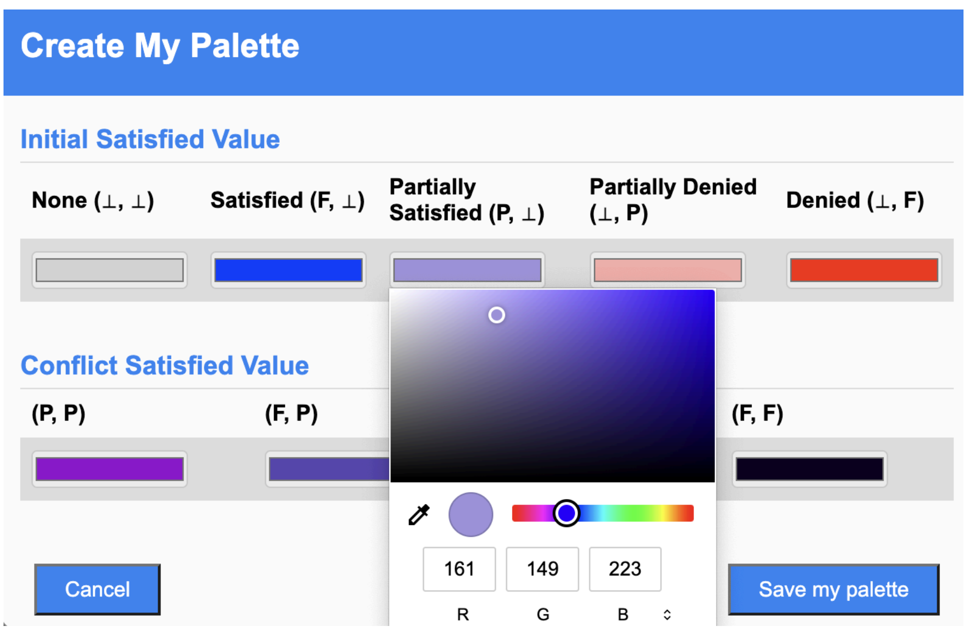


A color key for each palette is accessible via Documentation > EVO Color Key > [Palette of choice]. When activating the feature for the first time (by toggling the EVO slider in the top toolbar), the default selection is the red-blue palette, with its corresponding color key displayed.



My Palette:

Users can also customize their own palette by selecting the "Edit My Palette" option in the EVO dropdown menu, allowing them to adjust the color values for each satisfaction level.



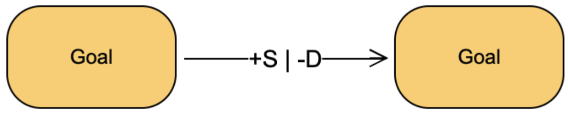
Evolving Functions

In Modeling mode, users can select an evolving function along with the initial satisfaction value. Below is a summary of evolving function options where FS refers to “Fully Satisfied” and FD refers to “Fully Denied”

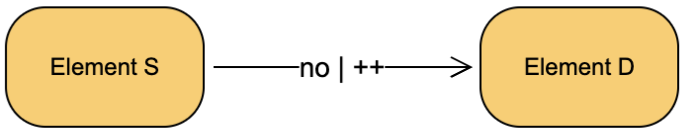
Elementary Functions	
Constant (C)	The satisfaction value remains constant at constantValue
Increase (I)	Changes in the satisfaction value become “more true” to a MaxValue as time progresses
Decrease (D)	Changes in the satisfaction value become “less true” to a MinValue as time progresses
Stochastic (R)	Changes in the satisfaction value are stochastic or random
General-Compound Functions	
User-Defined (UD)	Its function is a stepwise function defined by a sequence of other functions, repeating behavior can be specified over a subset of functions
Common Compound Functions	
Satisfied-Denied (SD)	The satisfaction evaluation remains FS until t_i and then remains FD
Denied-Satisfied (DS)	The satisfaction evaluation remains FD until t_i and then remains FS
Stochastic-Constant (RC)	Changes in the satisfaction value are stochastic or random until t_i and then remain constant at ConstantValue
Constant-Stochastic (CR)	The satisfaction values remains constant at ConstantValue until t_i and then changes in the satisfaction value are stochastic or random
Monotonic Positive (MP)	Changes in the satisfaction value become “more true” to a MaxValue at t_i and then remain constant at ConstantValue
Monotonic Negative (MN)	Changes in the satisfaction value become “less true” to a MinValue at t_i and then remain constant at ConstantValue

Evolving Relationships

Along with the constant relationships explained above, users can choose an evolving relationship model. Here, the link between nodes features a 'start' and 'end' relationship, incorporating previous decompositions and contributions.



The NO relationship: it indicates the absence of any connections and could be used in evolving relationships in addition to any previously mentioned relationship.



Assignments List

In “Modeling” mode, the top toolbar includes a “View Assignments List” button that enables users to create and/or modify the Max Absolute Time, Absolute Time Points, Relative Intention Assignments, Absolute Intention Assignments, Absolute Relationship Assignments, and Presence Condition Assignments.

Max Absolute Time:

It refers to the amount of time during which the model is representing the situation and is measured in arbitrary units. The Max Absolute Time is the maximum possible number of the timeline users can make.

Absolute Time Points:

They specify events, such as the change in the satisfaction level of an intention, that occur at specific time points during the overall range between Time 0 and the Max Absolute Time.

Relative Intention Assignments:

If the model has two or more intentions with Relative Time Points assigned by Evolving Functions, users can specify the relative order in which the Relative Time Points occur.

Absolute Intention Assignments:

If intentions have assigned Evolving Functions, users have the option to set the time point at which the satisfaction value changes.

Absolute Relationship Assignments:

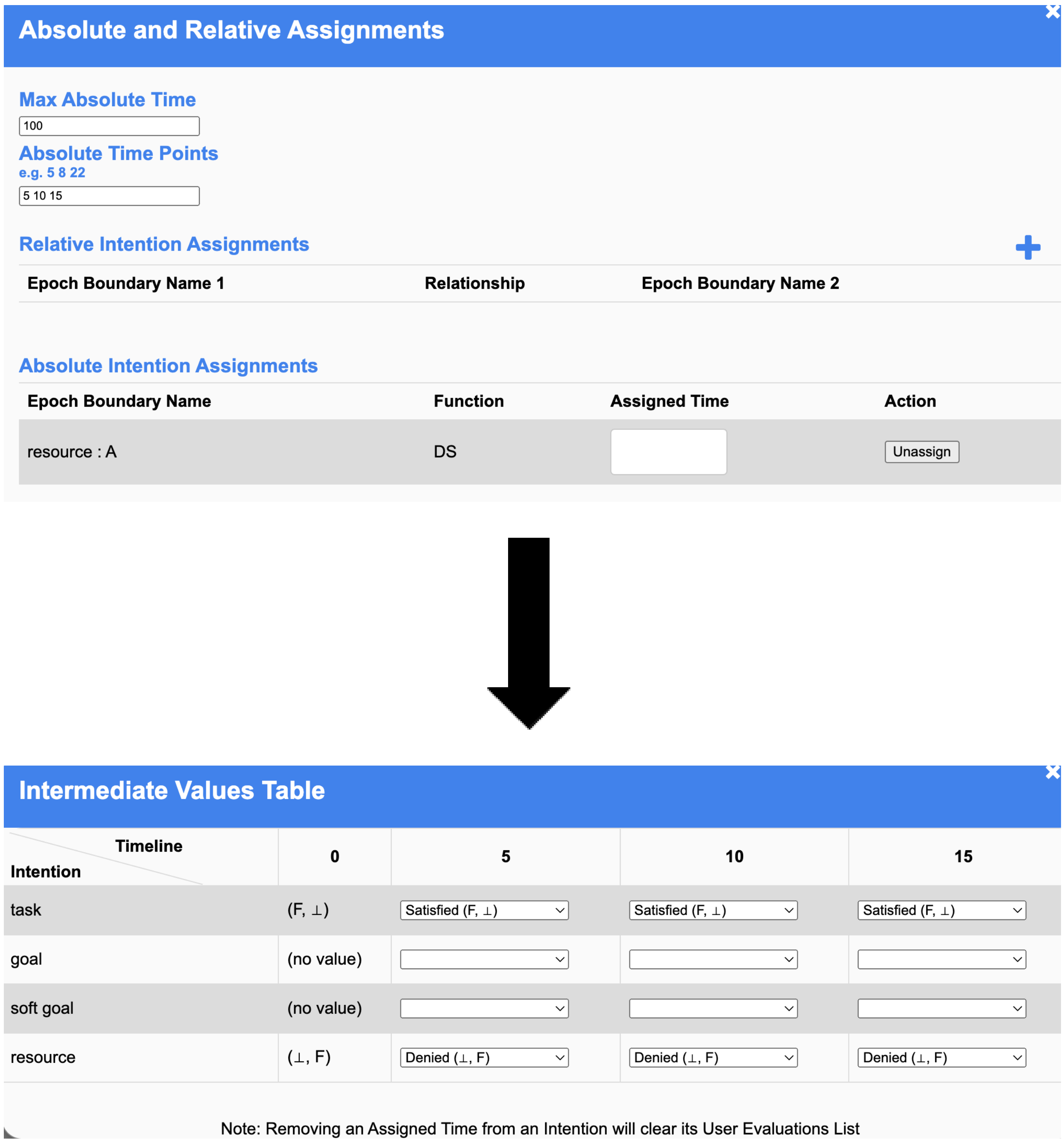
When a link in the model evolves over time, users have the option to set the time point at which the relationship changes.

Presence Condition Assignments:

Intentions and actors are present in the model throughout the time period by default. However, users can specify the presence intervals of any actor and/or intention by changing the “Available Interval” of the element.

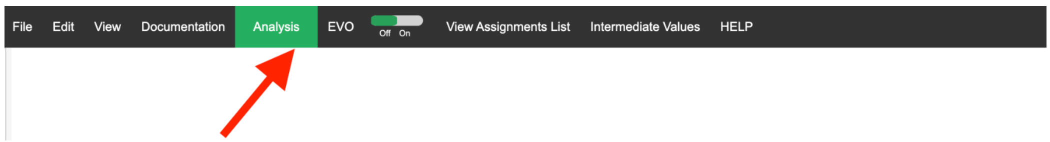
Intermediate Values

In the case where users have specified the Absolute Time Points and the model has intentions with Relative Time Points, the Intermediate Values Table can be used to specify where those Relative Time Points occur in relation to the absolute time points.



Model Analysis

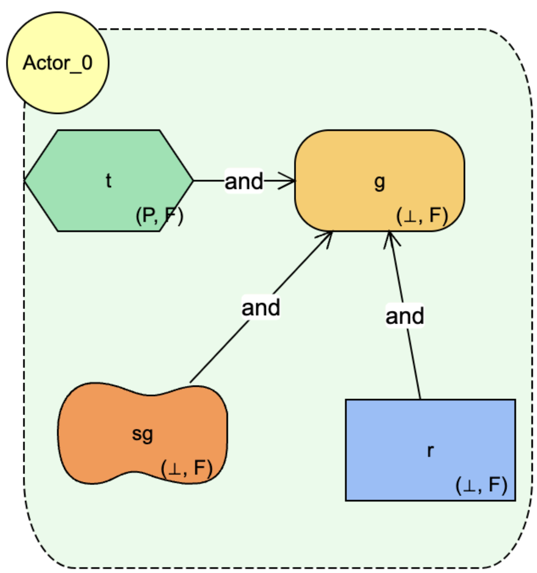
To switch between 'Modeling' and 'Analysis' modes, users simply click the green button located in the top toolbar. When in 'Analysis' mode, the button changes to red, indicating that users can return to 'Modeling' mode by clicking it again



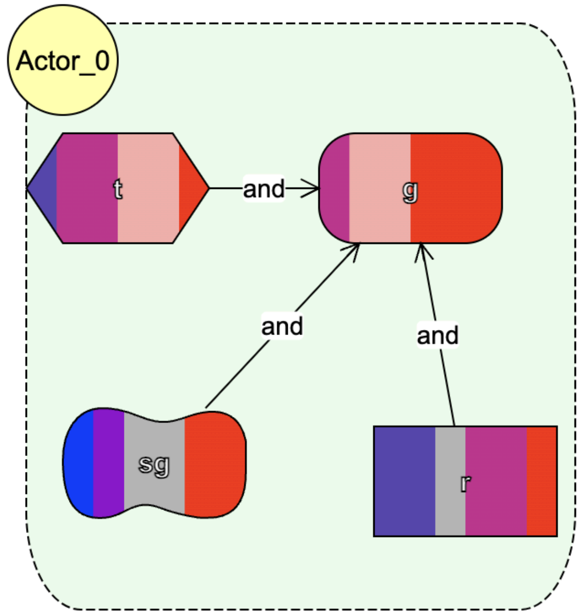
<u>Path Simulation</u>

Before clicking on 'Simulate Path,' users should set the analysis parameters, such as the conflict prevention level and the number of relative time points. Once the simulation is executed, users can then toggle the EVO slider in the top toolbar to one of the following stops:

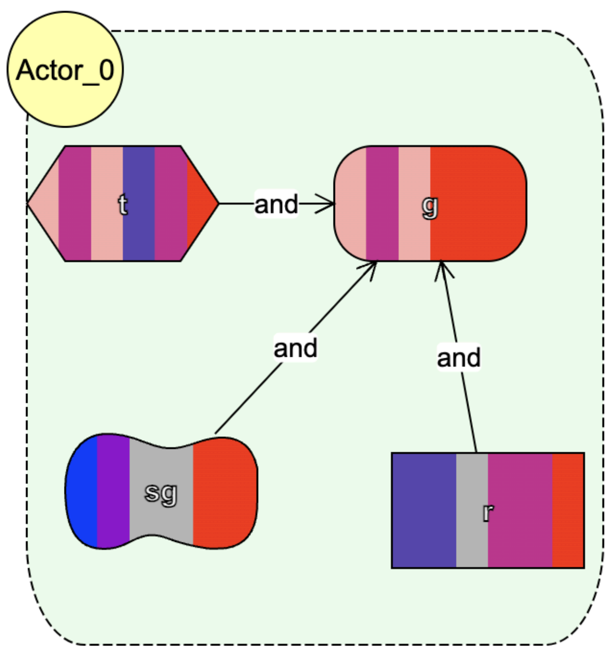
- OFF: This mode shows the evolution of the intentions as time progresses. Toggling the lower slider reveals the satisfaction value of each node at the given state.



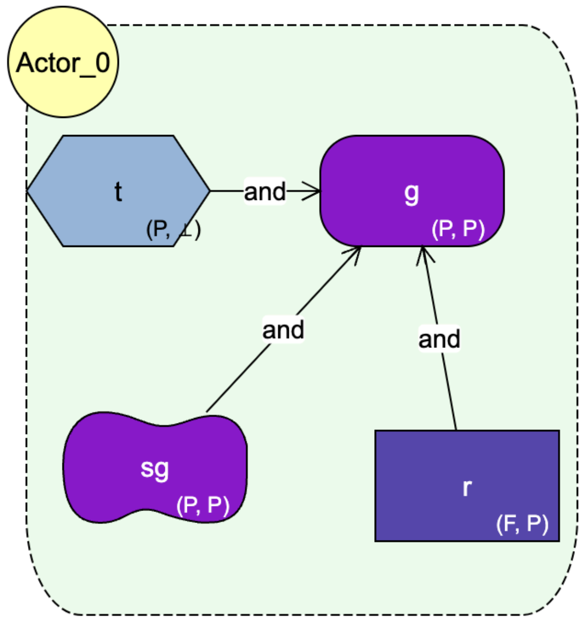
- %: This mode uses color-coding based on the selected palette and reveals the different satisfaction levels each node can take. Thicker bands represent more frequently appearing satisfaction values.



- Time: This mode uses color-coding based on the selected palette and reveals the change in satisfaction level for each node as time progresses.

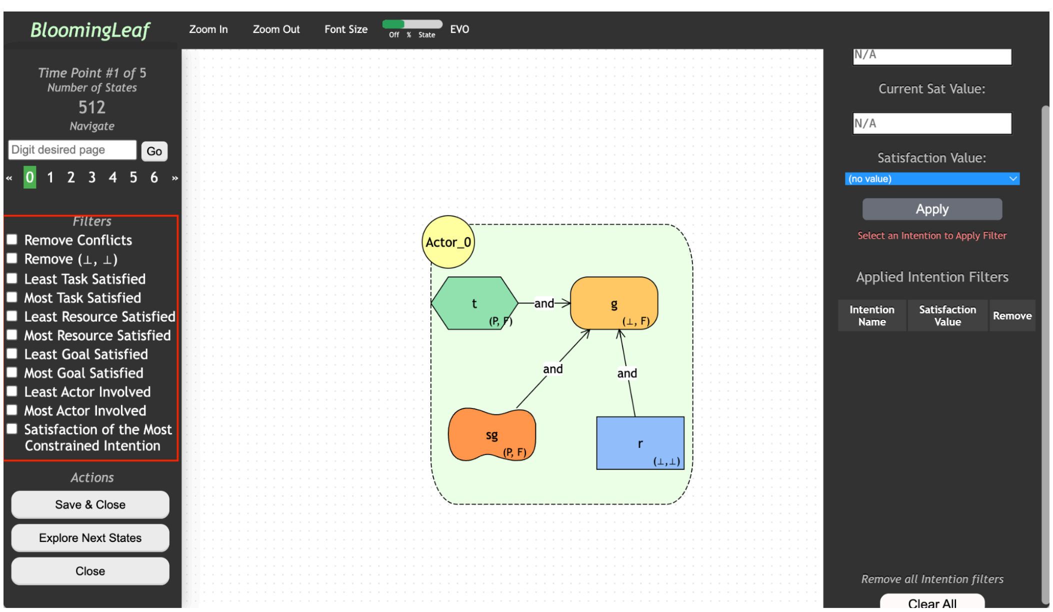


- State: It uses the same logic for OFF mode except each node in the model becomes color-coded based on the selected palette.



Next State:

This feature is only activated when the EVO slider is set to the “State” mode and when the state slider (the lower slider at the bottom of the page” is not set to the last time point. When clicking on the “Explore States” button on the top toolbar, a new window appears where users can apply any of the filters shown on the left panel:



Alternatively, users can apply intention filters based on the satisfaction value of the node.